Jiawei Qian

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PROFESSIONAL EXPERIENCE

AI Agent Engineer, Meituan

Sep 2025 – Present

End-to-end owner of AI Browser core intelligence and experience, responsible for agent quality iteration, deep user-scenario definition, core case design, and next-generation product capability evolution. Downward scope covers the Agent Harness self-iteration system, agent runtime architecture, and training flywheel; upward scope defines the product form, capability boundary, and experience baseline of Browser x Agent across browser-use / computer-use scenarios.

Methodology

- Designed the Agent Loop through a control-theoretic lens: treated the LLM policy as the plant, the harness as the controller, and evaluation / semantic observer as feedback sensors; explicitly separated setpoint, state, observation, and actuation to reduce oscillation and divergence in long-horizon tasks.
- Reframed agent-quality iteration as a training-style process: benchmark score becomes the objective, structured attribution from auto-analysis becomes the loss, and context structure, memory organization, Agent Skills, and system prompts become the updatable parameter space for gradient-like iteration.

Agent Harness Self-Iteration Pipeline (Core Direction)

- Designed and shipped the closed-loop auto-evaluation -> auto-analysis -> auto-iteration system: a higher-order agent controller evaluates sub-agent runs, attributes failures, decides iteration plans, and applies updates, upgrading agent improvement from human-driven tuning to system-driven iteration.
- Decomposed the full adjustable surface of agent quality into independently optimizable dimensions: context structure and denoising, tool-space definition and composition, system prompt engineering, Agent Skill injection and orchestration, and GUI Agent vs CLI Agent vs hybrid-mode routing.
- Built automatic Skill / Playbook distillation and injection: critical paths, UI elements, and action sequences are traced from execution trajectories, converted into reusable Agent Skills, injected at test time, and reused for self-evolution as a skill-level iteration flywheel.
- Designed the evaluation layer with rule-based checks and LLM-as-judge rubrics, combining task completion, trajectory efficiency, and behavioral compliance into structured attribution signals for auto-analysis rather than a simple pass rate.
- Designed the analysis and decision layer that maps evaluation signals to bottleneck dimensions such as missing context, tool gaps, prompt ambiguity, or skill gaps, then generates executable iteration proposals and applies them back into the harness.

User Case Mining & Agent Capability Definition

- User case mining & benchmark construction: mined measurable and regression-ready high-value task suites from real user journeys, including long-horizon information gathering, cross-tab / cross-site coordination, form and transaction operations, and complex web understanding, using them as agent setpoints and benchmark cases.
- Agent form & capability definition: defined the next-generation Browser x Agent interaction pattern, deep-scenario boundary, and experience baseline; mapped model upgrades, harness evolution, skill distillation, and context / memory mechanisms to concrete capability stages from usable to stable, predictable, and scalable.

Browser-use & Computer-use Agent Runtime

- Designed and iterated a ReAct-loop-based agent runtime covering state management, observation abstraction, tool-calling protocol, and failure recovery, improving execution stability for long-horizon browser tasks.
- Decoupled soft optimization from hard constraints by introducing the State Tracker and Semantic Observer, shifting robustness from model-dependent behavior to architecture-backed guarantees and reducing logical oscillation in complex tasks.

Training Flywheel

- Explored an SFT data flywheel based on real expert trajectories, built lightweight-model alternatives for selected vertical tasks, and validated the replacement potential of 7B / 30B models in high-frequency scenarios to reduce token cost and end-to-end latency.

Software Engineer, Microsoft

May 2024 – Sep 2025

Owned agent-driven analytics systems, LLM evaluation pipelines, and large-scale data infrastructure, covering agent workflows, code generation, memory / context optimization, and cost / performance optimization.

Agent Workflow, Code Generation & Memory Optimization

- Advanced analytics and engineering workflows from manual execution to AI-native workflows powered by agents, code generation, and validation, reducing human operations through task decomposition, automated checks, and iterative retry loops.
- Built data-analysis agents with GPT-4o, LangGraph, and OpenManus / MetaGPT, connecting User Query -> Agent -> data analysis and visualization, with support for RAG, domain knowledge graphs, and internal analytics tools.
- Iterated on a SQL -> DataFrame code-generation loop with sandbox validation and regeneration, improving the executability and reliability of generated analytics code.
- Applied MemGPT-style layered context management for external knowledge, chat history, and tool feedback, reducing token usage per task by 30% and improving performance on multi-turn analytical tasks.

LLM Evaluation & Data Flywheel Infrastructure

- Built Azure ML evaluation pipelines with GPT-4o, o1, and DeepSeek V3 / R1, combining user signals to generate long-term profiles and automatically evaluate and monitor ranking and feed relevance for MSN / Bing scenarios at about 416M tokens/day.
- Designed cache and parallelization strategies for embedding and text-processing pipelines, using a hybrid LRU / LFU cache mechanism to achieve a 95%+ hit rate, reduce quota usage by 90%, cut component runtime by 60%, and improve pipeline robustness.
- Continuously optimized prompts and evaluation metrics on top of a golden set, improving precision and recall in relevance evaluation and supporting a closed loop for data labeling and iteration.

Multi-modal LLM Pipeline & Data Infrastructure

- Built multi-model content-understanding and monitoring pipelines for text, image, and video, supporting feed-quality analysis in recommendation scenarios across dynamic prompting, cache reuse, and pipeline optimization.
- Developed large-scale data pipelines with ClickHouse and Spark to aggregate user, content, and request-signal data; improved parallelism by restructuring connection logic and dependencies, reducing the core SLA from 5 days to 4 days while adding input / output monitoring to improve data stability.

AI Algorithm Engineer, Huawei

July 2022 – May 2024

Worked on training, modeling, and complex optimization problems, focusing on sequence modeling, model compression, and approximation optimization, providing a training and algorithmic foundation for later Agent / LLM systems work.

Modeling & Training

- Researched GRU / Transformer-based sequence modeling and entropy coding methods, using temporal probability modeling for NLP data compression and vector quantization for model-parameter compression, improving coding efficiency from 6 bits to 5 bits and increasing compression ratio by about 20%.
- Contributed to modeling and training of end-to-end deep learning communication systems; designed loss functions based on real phase-noise and power-amplifier distortion data and trained neural transceivers whose transmission performance approached the theoretical optimum under AWGN.

Optimization & Approximation

- Designed parallel approximate solvers for NP-hard resource allocation problems using randomized algorithms, greedy strategies, and selection mechanisms, then ensembled results with bagging / expert-style methods to reach about 95% of near-optimal integer programming quality while reducing solve time from 1500 seconds to 3 seconds.

EDUCATION

University of Waterloo

2021-2022

MEng, Electrical & Computer Engineering

Chongqing University

2015-2020

BEng, Electrical Engineering and Automation

SKILLS

Languages: Python, Golang, SQL
AI-native Engineering: Problem scoping & prioritization, AI task decomposition, human-in-the-loop workflow redesign, Cursor, Codex
LLM / Agent: Agent harness self-iteration pipeline, auto-evaluation / auto-analysis / auto-iteration, control-theoretic agent loop design (plant / controller / observer / feedback), training-style iteration (benchmark→loss→gradient over context / memory / skill / system prompt), browser-use / computer-use agents, ReAct, context engineering, tool orchestration, system prompt engineering, skill injection & design, LangGraph, RAG, prompt engineering
Training / Modeling: SFT, trajectory data construction, Transformers, GRU, CNN, model compression, offline evaluation, approximation optimization
Data / Infrastructure: Spark, ClickHouse, Azure ML, MySQL, Kafka, CI/CD, RESTful APIs
Language: English, TOEFL 104, GRE 330

PUBLICATION

Transformer-based Fuzzer   EAI MONAMI 2022

AWARDS

- First Prize, Chinese Mathematics Competitions for College Students
- Top 5 Coder, Huawei ICT Department (Python and Java)
- Huawei Future Star